

Session 4 · Workflows on Live SEC Data

Yesterday you pasted Apple's data by hand. Today your code fetches it from the U.S. Securities and Exchange Commission (SEC) — and screens the whole peer universe.

Plan: concepts 15' · live demo 22' · your lab 30' · debrief 8'

By the end of Session 4 you can

1. **Draw** the workflow pattern and defend the code-vs-model split
2. **Pull** real fundamentals from SEC filings with one function call
3. **Build** a screen you can rerun tomorrow, criteria as dials
4. **Audit** the model's own prose: every number traced to an input
5. **Train** a forecasting model and measure its error before trusting it
6. **Express** the whole pipeline as a LangGraph workflow graph

What we are doing, and why

- **Yesterday's valuation was one analysis, done once.** Today it becomes a **tool**: code that rebuilds the data, applies your criteria, and can be rerun any morning.
- **A workflow is a plan the model cannot change.** Code retrieves and filters; the model reasons once, in the middle, about a table it is given; code audits the prose; a human approves.
- **The data is primary-source:** audited filings from the SEC, fetched live — with the messiness of real data as part of the curriculum.
- **The finished pipeline gets the industry's name for it:** a LangGraph workflow graph, the standard way professional teams write these systems.

The storyline of this session

Apple, live from its filings



16 companies fetched



the deterministic screen



grounded rationales, audited



a trained forecaster



the pipeline as a graph

the Session 2 company, now from the source

Apple's peer group + pharma + industrials

criteria as dials – does Apple pass?

the model writes; code traces every number

its measured error is the product

LangGraph: the same steps, made explicit

A workflow is a plan the model cannot change

INPUT → RETRIEVE → STRUCTURE → REASON → VALIDATE → HUMAN
(code) (code) (model) (code) (you)

- **Code does the arithmetic:** pandas is deterministic; the model is not
- **The model reasons once,** in the middle, about a table it is *given*
- **Code audits the prose,** and a human approves: nothing is published without explicit sign-off

EDGAR: every filing, free, no key

- **EDGAR** (Electronic Data Gathering, Analysis and Retrieval) is the U.S. Securities and Exchange Commission's public filing database
- **What it contains:** every 10-K, 10-Q and 8-K since the nineties, machine readable
- **What it does not contain:** market prices and analyst estimates — filings are not market data

Three traps you will meet, because **companies tag their own accounts:**

1. The same item **changes names** across years
2. Foreign filers report in **local currency**
3. Most pharma reports **no operating income line at all**

Demo: a memo with two moments to watch

A company memo built live from the filings.

- **First, the data gaps:** the memo lists what it could *not* know from its inputs — the blind spots are declared, not hidden
- **Second, a planted error:** one figure in the memo is altered by hand, and the audit finds it

```
audit: every figure in the prose → checked against inputs → the altered  
figure is flagged
```

The libraries in this session

Library	Role here	Standing
<code>requests</code>	fetches SEC filings over HTTP	the standard HTTP library
<code>pandas</code>	the deterministic screen	the industry standard for data work
<code>numpy</code>	trains the least-squares forecaster	the numerical foundation of Python
<code>pydantic</code>	typed, validated model output	the industry standard for validation
LangGraph	the pipeline as an explicit graph	the industry standard for AI workflows, from the LangChain ecosystem

How the labs work

Every exercise sits between two markers. **You fill the gaps. Nothing else changes.**

```
### START CODE HERE ###  
mask = (df[None] >= min_growth) & (df[None] >= min_margin)  
### END CODE HERE ###
```

- **None** → replace with the correct column, value or variable
- **[QUESTION IN CAPITALS]** → replace with the text the bracket asks for
- **Everything else is given.** Do not rewrite it.
- Then run the  **check cell** directly below. Green means correct: continue.

Stuck for two minutes? Select the lines, press **Option+K** (**Alt+K** on Windows), and ask the ***** panel.

Your lab · notebook 04 · 30 minutes

A screening engine: Apple's peer universe + pharma + industrials, live from EDGAR

- **Exercise 1:** fetch 3 years of fundamentals for 16 companies, tolerating failed tickers
- **Exercise 2:** the filter — **pandas decides which companies pass, never the model.** Watch for what happens to Apple
- **Exercise 3:** one grounded rationale per survivor + numeric audit
- **Exercise 4:** train a revenue forecaster; the measured error is the product
- **Exercise 5:** wire the same pipeline as a LangGraph graph — it draws itself

Two questions for the debrief: why net margin, not operating margin?

And: does Apple pass its own screen — and what does the answer mean?

Key takeaway

A screen you can rerun tomorrow is worth more than ten analyses performed once.

Next session: the model plans its own steps, under controls you define —
and your finished tool goes on GitHub.